

- 5 a** free: oscillation without any loss of energy / no external forces e.g. resistive forces B1
- forced: body is made to vibrate by an (external) periodic driving force such that continuous energy is input B1
- b** From the graph, maximum displacement = 23.00 cm A1
 minimum displacement = 11.00 cm
 amplitude = 0.5 (23.00 – 11.00)
 = 6.00 cm = 0.0600 m
- c** From graph, A1
 T = 2.000 s [M1]

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{2.000} = 3.14 \text{ rad s}^{-1}$$
- d** Energy of Oscillation = Maximum KE M1

$$= \frac{1}{2} m \omega^2 x_0^2 = \frac{1}{2} (10) (3.14)^2 (0.06)^2 = 0.177 \text{ J}$$
 A1
- e** For loss of contact, $a > g$
 At this setting, $a_{\min} = 9.81 = 3.14^2 (x_{\min})$ M1
 $x_{\min} = 0.995 \text{ m}$ A1

